# Draft SEDI ACDCs
# Utah: State Endorsed Digital Identity

*Samuel M. Smith PhD*
*Keri Foundation Co-Chair*
*sam@keri.foundation*




*https://keri.foundation*

# What is SEDI?

Opt-in Citizen-Sourced State-Endorsed Identity

https://sedi.utah.gov/

Constitutional Approach that Protects Citizen Rights and Liberty

Restrictions on Government Action

Duty of Data Loyalty

Enabling Legislation 2026

Enforcement Legislation 2027

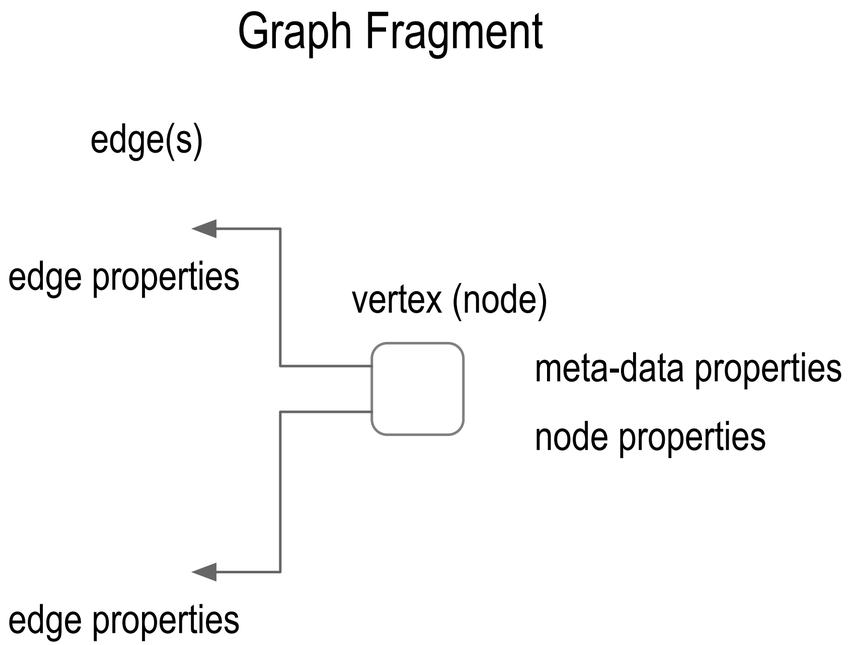
Multi-State Consortium

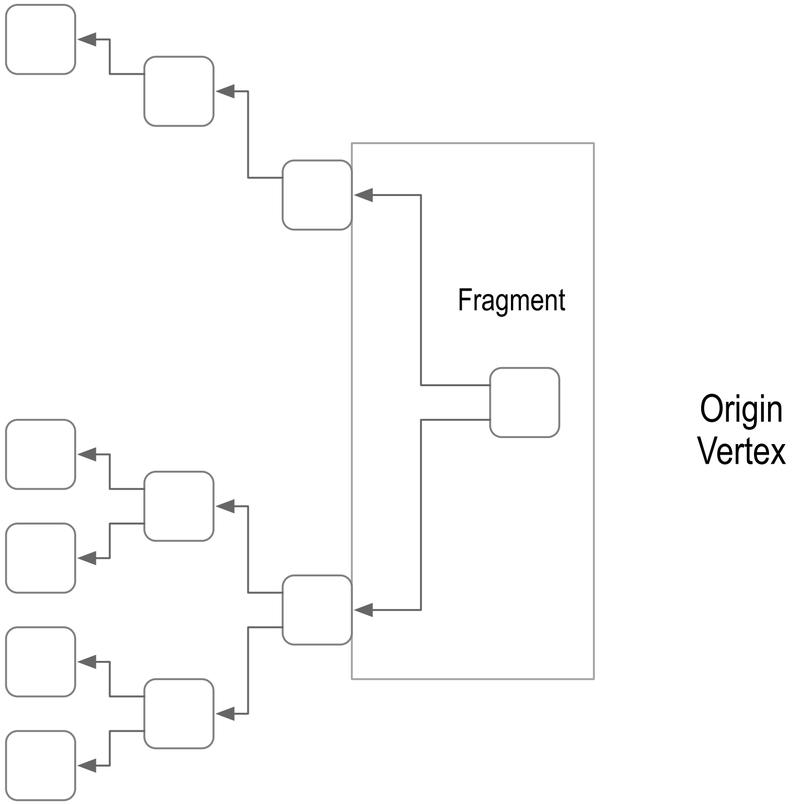
Fall SEDI Summit Park City Utah Nov 17-19

https://www.sedisummit.com/



# Using ACDCs




Perpetually Verifiable Issuances

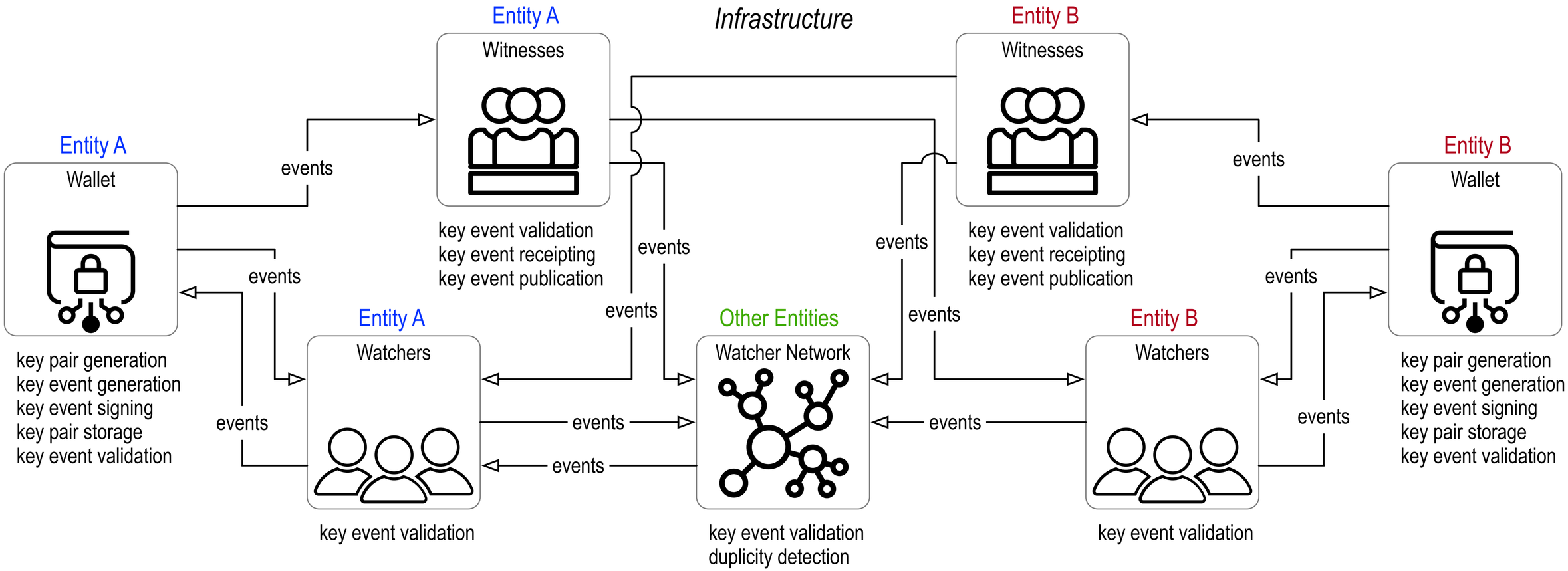
Each Fragment is Separably Authenticatable

Authority: (DAAART)

Delegable Attenuable Aggrable Auditable Revocable Transferable

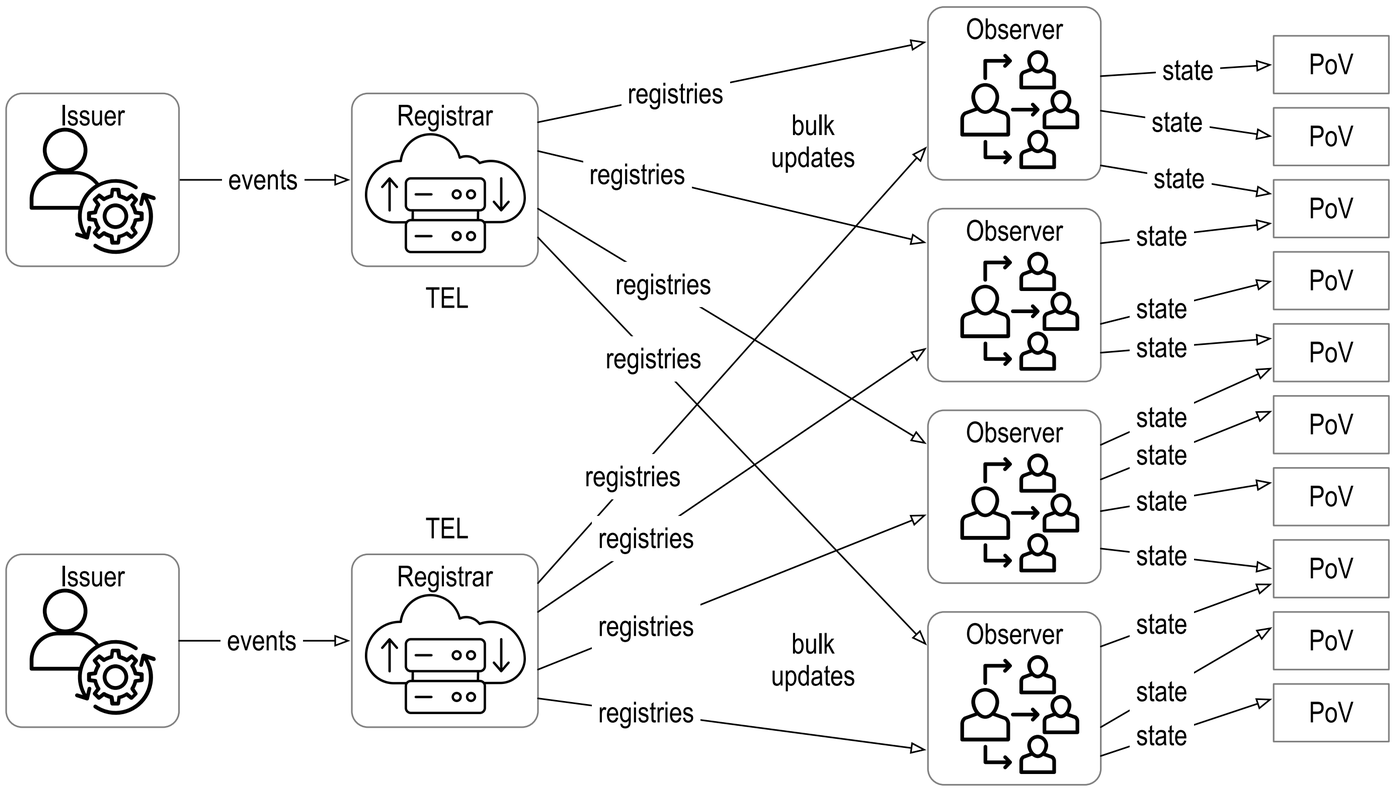
High Fidelity Modeling of Real World Authority Structures

# KERI Infrastucture



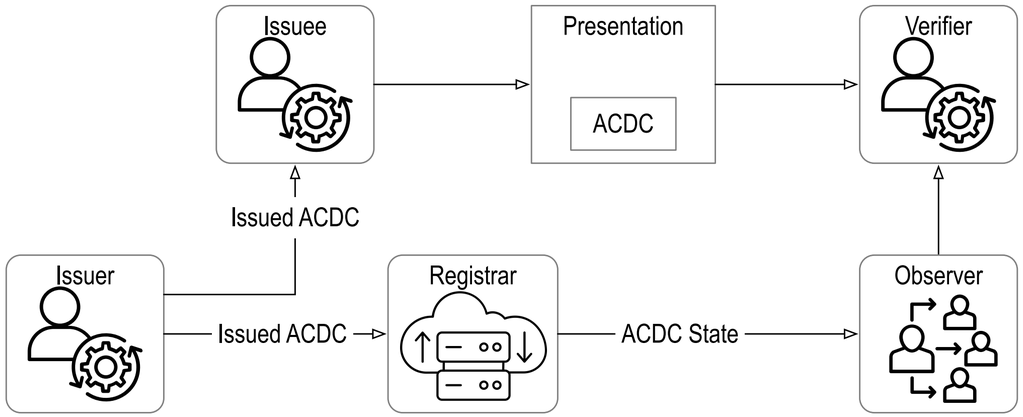
# KERI Infrastucture ...

## SEDI Bulk Issuance

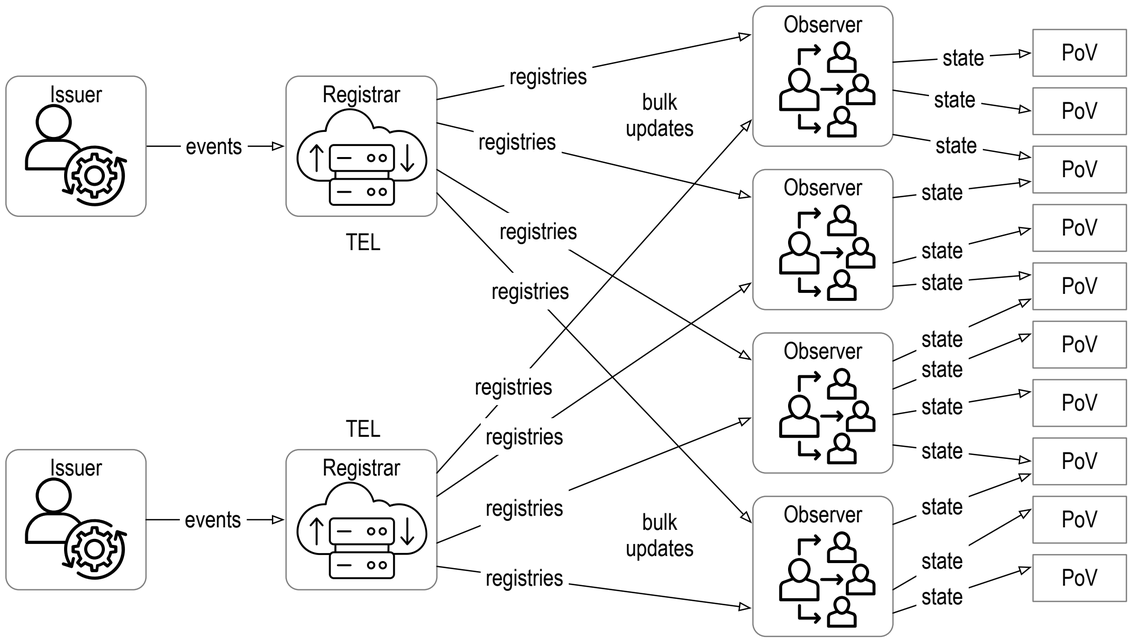


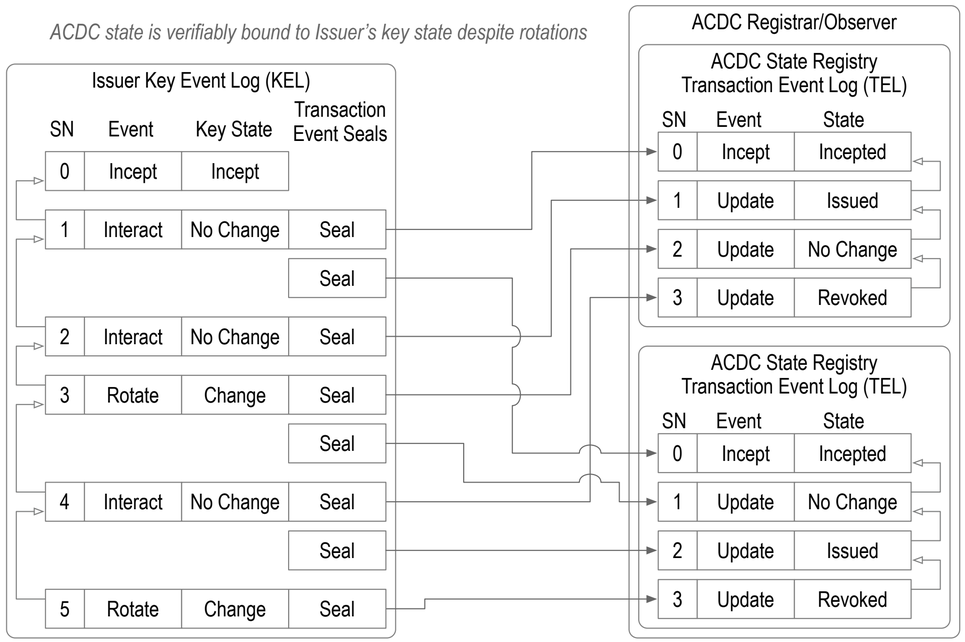

## Sparse Merkle Trees

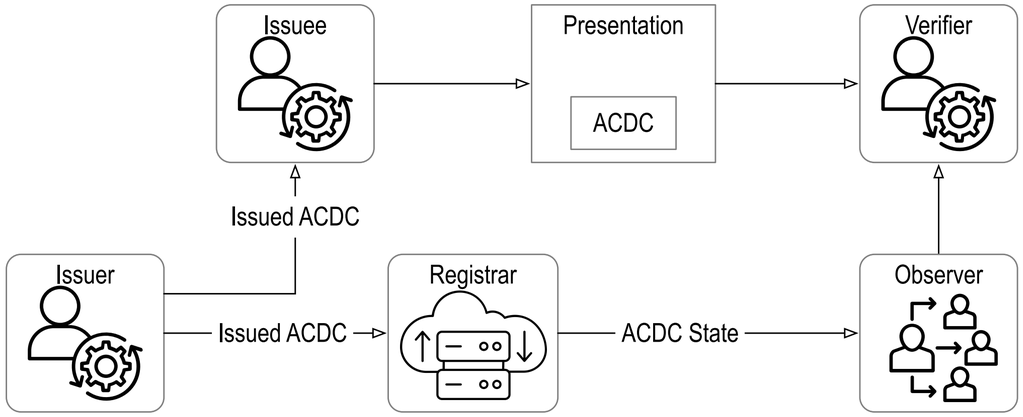
## Issuance Verification at Presentation



# Issuance Registry - ACDC State




## Blindable Registry = Perfect Forward Privacy

ACDC

| v | t | d | u | i | rd | s | a | e | r |
|---|---|---|---|---|---|---|---|---|---|
| Version | Type | SAID | UUID | Issuer AID | Registry | Schema | Attribute | Edge | Rule |

Registry Inception Event (rip)

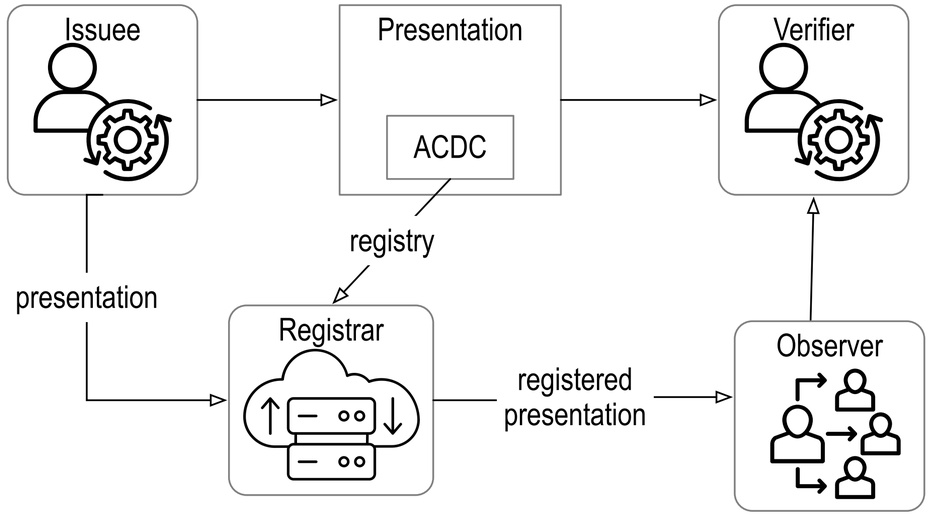
| v | t | d | u | i | n | dt |
|---|---|---|---|---|---|---|
| Version | Type | SAID | UUID | Issuer AID | SN | DateTime |

Registry Blinded Update Event (bup)

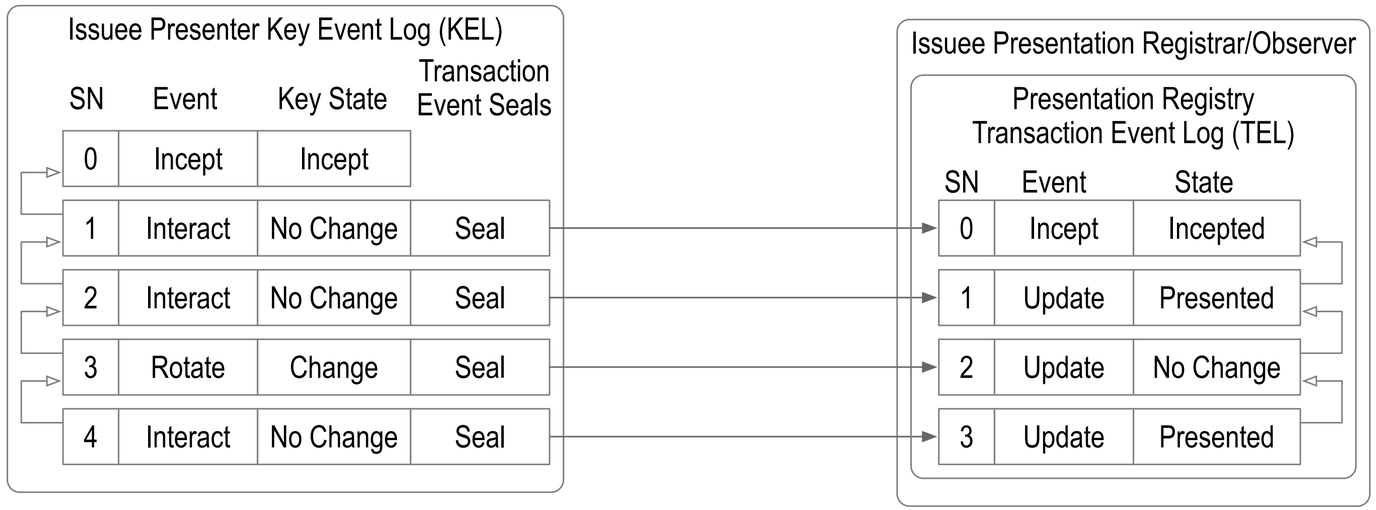
| v | t | d | rd | n | p | dt | b |
|---|---|---|---|---|---|---|---|
| Version | Type | SAID | Registry | SN | Prior | DateTime | BLID |

# User Fraud Presention Detection

## User Presentation Registries



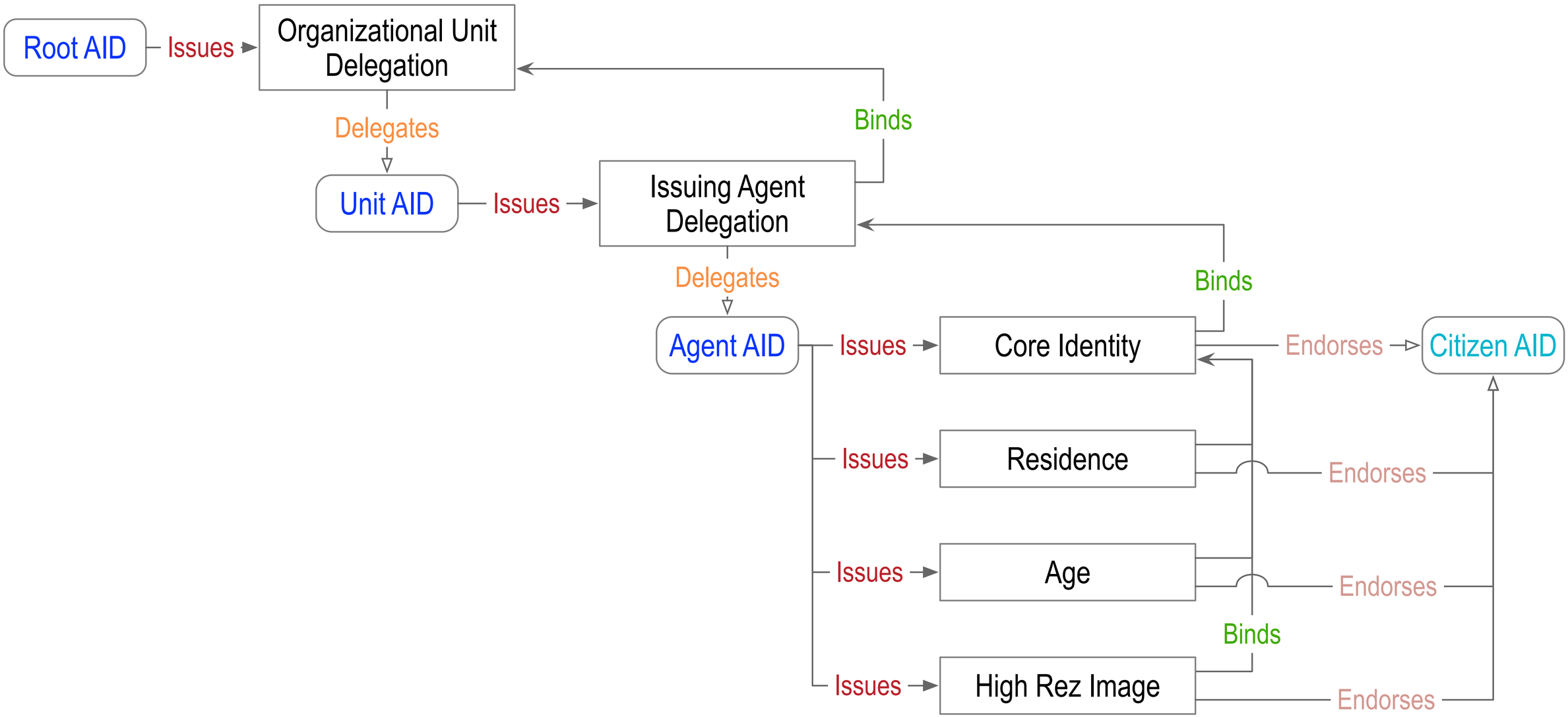

*ACDC Issuee Presentation state is verifiably bound to Issuee's key state despite rotations*



# SEDI Implementation Guide Draft

*https://github.com/keri-foundation/sedi/blob/main/sedi_guide.md*

# SEDI: Delegation to Citizen Core Entitlements



# SEDI: Delegation Org Unit


```
{
  'v': 'ACDCCAACAAJSONAAIn.',
  't': 'acm',
  'd': 'EIojbncOpuCoMFWZSutjfkLf5KNu_nbCOsqgFClk3SMK',
  'u': '0AAxOP-l4O8GZPbv7dQBeNfM',
  'i': 'EAN4gwVu_a9EDxFn-camF6OoQvZAvi-_FyluyyZFKzs5',
  'rd': 'EA3Y6LeoyNFjnLS1xZoRQnzX0fWgUw0XjD4IgIf8ZHxD',
  's': 'EBbOpBe0lP_epHhakeiXdOmeal3leDfLRf0PAItsanoQ',
  'a':
  {
    'd': 'EHydQsy-4di85pJaiPqVN9_APpLtU15QXcmVFEN2FQE-',
    'u': '0AAk4UxXJc8mbqWSdtAPkK6w',
    'i': 'EMrUfTuv8j3vBVAgPSlDi1D_o35F5uwsIAWjBatLhK9E',
    'issuedDate': '2020-08-01T00:00:00.000000+00:00',
    'unit': 'SediProgramOffice'
  },
  'r':
  {
    'd': 'EFPxq4WPl29szUqbrQIviOh_Ls_RlrYbp4L-fdQH0XrX',
    'l': ''
  }
}
```

# SEDI: Delegation Issuing Agent


```
{
  'v': 'ACDCCAACAAJSONAAO9.',
  't': 'acm',
  'd': 'EOD4QiWo1oa6UtiEr456Lu0ba3v1Cmd6qOPoRbAdx2hD',
  'u': '0ACpkR3a0WoJk8BHz4M6aZVo',
  'i': 'EMrUfTuv8j3vBVAgPSlDi1D_o35F5uwsIAWjBatLhK9E',
  'rd': 'EEy3daQxc9NrgzA1V7KjgOqLF_te2gs2-sYElTHsPzYE',
  's': 'EGx4BLclkjhaK1501guyBifxuTTJLwuK61InBTdkKF7v',
  'a':
  {
    'd': 'EFPQAD6XLlU9bnNPMWfjl1Q-Rf20Jwoip_1lkp8XGKKt',
    'u': '0ADtdKN8DlnUKUFDN47QKfNj',
    'i': 'EKBCU6u_xObNhFc9uuz1VdntNt99xmB2fA5qz7Li-Sl-',
    'issuedDate': '2020-08-01T00:00:00.000000+00:00',
    'role': 'SediIssuingAgent',
    'name':
    {
      'd':
'EJj1BPhB8uGFLA8QtNjNJ9MXJ8fvOliIWj5w7YLzReWI',
      'u': '0AC0eUUAKHVXdJMDVXtx1yS3',
      'value': 'Susan Park'
    }
  },
  'e':
  {
    'd':
'EIZC3vk0dgDvZNV3Vf1z3mav0HkMbGHq0vxm7NcbC6WP',
    'u': '0AB4w_WXy8JmoUebF1aEnOD_',
    'orgUnit':
    {
      'd':
'ENZvB2t1Tit1Z2PSSebS70IKEq33JynJrWK-rjR_cSMA',
      'u': '0ABl8aVsQRgBVSWnMQcwgDQ7',
      'n':
'EIojbncOpuCoMFWZSutjfkLf5KNu_nbCOsqgFClk3SMK',
      's':
'EBbOpBe0lP_epHhakeiXdOmeal3leDfLRf0PAItsanoQ',
      'o': 'DI2I'
    }
  },
  'r':
  {
    'd': 'EFPxq4WPl29szUqbrQIviOh_Ls_RlrYbp4L-
fdQH0XrX',
    'l': ''
  }
}
```

# SEDI: Core Identity

```
{
  'v': 'ACDCCAACAAJSONAAfN.',
  't': 'acm',
  'd': 'EGsPyGCyHtDSWP61rVsRh3F8qooP_jrX6qPHqHn7-FOg',
  'u': '0ABQNZNkD1y0W4mglDB4ei5Y',
  'i': 'EKBCU6u_xObNhFc9uuz1VdntNt99xmB2fA5qz7Li-Sl-',
  'rd': 'EDOfxmEeOsWdi5ZQyuy98W4s15vmV1RWVFuqm4GZflTn',
```

```
'a':
{
  'd': 'EPIuMf4tkzdqVMSuq59rBD14LZpmFtFjhnh8Io1BWdbt',
  'u': '0ACd8yXBMGBLNDwr-MMgtris',
  'i': 'EDB8gKNwzurf33pV2hsyGR9XFOmitDhc0LUzDamcU2JR',
  'rd': 'ELUW5D0X0pMFM30ZHZKB997lad86PISushrKkKzlhQrl',
  'primary': True,
  'givenName':
  {
    'd': 'EBu_EdUcstZq6woZ7NMe2pyU7jjcQOdC9w1ryHa1P_Sq',
    'u': '0ADGYtYEzEdGpaq_sDXwamDm',
    'value': 'Guy'
  },
  'middleName':
  {
    'd': 'EOhalIHhb5ZbrJZ5SMY_vWBj2ds_z9W8mJ3j-FTjgSUz',
    'u': '0ACEZIR6pk97xr2cy-gBdod4',
    'value': 'Marty McFly'
  },
  'familyName':
  {
    'd': 'EDBg78wYuNEQkcv-XciKFkxDQmRvIeTsXfjVMJLtUjYW',
    'u': '0ACUdqI4OVtXDL5BBO13QdrJ',
    'value': 'Brown'
  },
 'nameSuffix':
  {
    'd': 'EFDPwXKE-3wg-WTUVu4GWfMeu4bj8rGNkvdFrMo4Ja4N',
    'u': '0ACXabyEAzJ1U-4qOek3adv5',
    'value': 'Jr'
  },
```

# SEDI: Core Identity ...

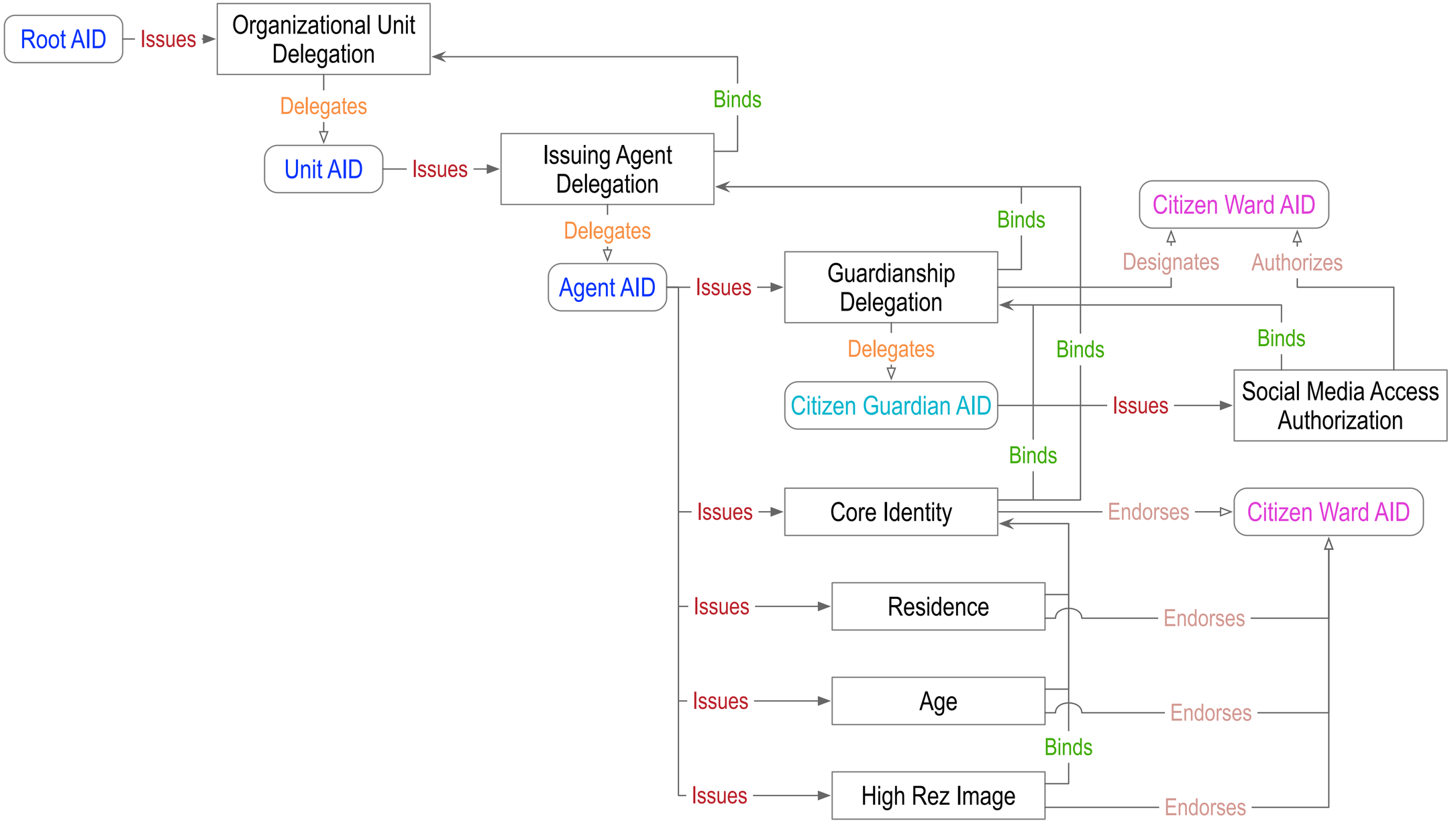
```
  'birthDate':
  {
    'd': 'EMGKn6dwPJMd79vWaGEv7OlCQQ0oGd0nt0fRqBfz1ZyA',
    'u': '0ABR4UtpgSSmCSVF1eNJ6joz',
    'value': '2002-08-22T00:00:00.000000+00:00'
  },
  'facialImageProof':
  {
    'd': 'EInFQsE3pfRWQp3KrpH91f7YMCR2wslrTOTKL5SF6_E7',
    'u': '0ABqxMB1vXX4RL4tFUOLzn8y',
    'value': 'EIQw_2CqmmC96YYUFXTW8XSkLQU2-v9bDCyItazmKhTW'
  },
  'legalPresenceStatus':
  {
    'd': 'EMfOp3-Dd1ZXSDUHNWE4ohAsU0Qp8tfQV_gUYKSJE7-Q',
    'u': '0AAvTJo84OMiipKf_90mn7bZ',
    'value': 'citizen'
  },
  'issuedDate':
  {
    'd': 'EFkQuDu6twqFQ3GdSpK-jxKRYsxxCx-iSbh9RMbTbIsE',
    'u': '0ACqPN2zhcII6TVGRKYC1ckJ',
    'value': '2020-08-22T00:00:00.000000+00:00'
  },
  'expirationDate':
  {
    'd': 'EOjYb0hwGBe1rA0Ui8foLMoCu5lItuhnodomghuyY4r9',
    'u': '0AAfMYErqhjBCf4cgG5huzT2',
    'value': '2028-09-01T00:00:00.000000+00:00'
  }
},
```

```
  'e':
   {
     'd': 'EFwdv1MuJAVYnyuhee0kAjDIpF44oFn5oFo9xoVwtId1',
     'u': '0AADjz761fupJiCvj4GWFBfg',
     'utahAgent':
     {
       'd': 'EIh9Bz0kGwGyfDbDe42W1zA9Ukj2DuH-95eebY57ctI6',
       'u': '0AAsHaHfgfNEaRuRwI1Y77ry',
       'n': 'EOD4QiWo1oa6UtiEr456Lu0ba3v1Cmd6qOPoRbAdx2hD',
       's': 'EGx4BLclkjhaK1501guyBifxuTTJLwuK61InBTdkKF7v',
       'o': 'DI2I'
     }
   },
   'r':
   {
     'd': 'EFPxq4WPl29szUqbrQIviOh_Ls_RlrYbp4L-fdQH0XrX',
'l': ''
   }
}
```

# SEDI: Core Identity Partial Disclosure

```
 a':
{
  'd': 'EPIuMf4tkzdqVMSuq59rBD14LZpmFtFjhnh8Io1BWdbt',
  'u': '0ACd8yXBMGBLNDwr-MMgtris',
  'i': 'EDB8gKNwzurf33pV2hsyGR9XFOmitDhc0LUzDamcU2JR',
  'rd': 'ELUW5D0X0pMFM30ZHZKB997lad86PISushrKkKzlhQrl',
  'primary': True,
  'givenName': 'EBu_EdUcstZq6woZ7NMe2pyU7jjcQOdC9w1ryHa1P_Sq',
  'middleName': 'EOhalIHhb5ZbrJZ5SMY_vWBj2ds_z9W8mJ3j-FTjgSUz',
  'familyName': 'EDBg78wYuNEQkcv-XciKFkxDQmRvIeTsXfjVMJLtUjYW',
  'nameSuffix': 'EFDPwXKE-3wg-WTUVu4GWfMeu4bj8rGNkvdFrMo4Ja4N',
  'birthDate': 'EMGKn6dwPJMd79vWaGEv7OlCQQ0oGd0nt0fRqBfz1ZyA',
  'facialImageProof': 'EInFQsE3pfRWQp3KrpH91f7YMCR2wslrTOTKL5SF6_E7',
  'legalPresenceStatus':
  {
    'd': 'EMfOp3-Dd1ZXSDUHNWE4ohAsU0Qp8tfQV_gUYKSJE7-Q',
    'u': '0AAvTJo84OMiipKf_90mn7bZ',
    'value': 'citizen'
  },
  'issuedDate': 'EFkQuDu6twqFQ3GdSpK-jxKRYsxxCx-iSbh9RMbTbIsE',
  'expirationDate':
  {
    'd': 'EOjYb0hwGBe1rA0Ui8foLMoCu5lItuhnodomghuyY4r9',
    'u': '0AAfMYErqhjBCf4cgG5huzT2',
    'value': '2028-09-01T00:00:00.000000+00:00'
  }
}
```

# SEDI: Guardianship



# SEDI: Guardian Delegation

```
{
  'v': 'ACDCCAACAAJSONAASb.',
  't': 'acm',
  'd': 'EONzgE0pcYF7hz1E_VVN4ea9iv75Q_C2ifhsZV1j6uQp',
  'u': '0ADXSX677aGYvS4d5a1u6J2X',
  'i': 'EKBCU6u_xObNhFc9uuz1VdntNt99xmB2fA5qz7Li-Sl-',
  'rd': 'EDOfxmEeOsWdi5ZQyuy98W4s15vmV1RWVFuqm4GZflTn',
  's': 'EDJzXg34opr6-odFix6lpiccfHF7ksITHuIES_MQRH4b',
  'a':
  {
    'd': 'EDhdyFmwvPUkniyoHMNDQOpk_cJsN_FYPEUqiklW4NMs',
    'u': '0ADk_KpjHu2tltJJrAOUDSSQ',
    'i': 'EDB8gKNwzurf33pV2hsyGR9XFOmitDhc0LUzDamcU2JR',
    'rd': 'ELUW5D0X0pMFM30ZHZKB997lad86PISushrKkKzlhQrl',
    'role': 'parent',
    'ward': 'EKr8JLtfqWCmHrxO3yu8ocS2n9o0Tlspeaqm9ZOf3FM1',
    'issuedDate':
    {
      'd': 'EHDyi3drPTNJ6WDAvPIbcOqtRV03mTPLRuRyTSzQUfpS',
      'u': '0AALcJnxH3SlR-HiMZ6evv0U',
      'value': '2012-06-21T00:00:00.000000+00:00'
    },
    'expirationDate':
    {
      'd': 'EPtJr5V6lRCBvT4QudVp7ucFspyjWPRhedf9MRG_CaEa',
      'u': '0ACbGWBFSYSV2WYripA_fQ2S',
      'value': '2030-06-21T00:00:00.000000+00:00'
    }
  },
  'e':
  {
    'd': 'EJLcbUvhJovz0feXaG5qddR5cvlsLk2KzFCtu56JzvsP',
    'u': '0ACbGWBFSYSV2WYripA_fQ2S',
    'utahAgent':
    {
      'd': 'EI9FLb6B7Oe54SW6A1Rk_PMj3zHe7Cc3Ukup97SJR5X8',
      'u': '0ABgAUzqg-7DB3Mn6Uqoiyla',
      'n': 'EOD4QiWo1oa6UtiEr456Lu0ba3v1Cmd6qOPoRbAdx2hD',
      's': 'EGx4BLclkjhaK1501guyBifxuTTJLwuK61InBTdkKF7v',
      'o': 'DI2I'
    }
  },
  'r':
  {
    'd': 'EFPxq4WPl29szUqbrQIviOh_Ls_RlrYbp4L-fdQH0XrX',
    'l': ''
  }
}
```

# SEDI: Ward Social Access Authorization


```
{
  'v': 'ACDCCAACAAJSONAAOu.',
  't': 'acm',
  'd': 'EGdxe5u7SRt__mJBvjRc0LdJycRDYgdnrjwdce9UQDE6',
  'u': '0ADxmNUC42qQYJNlAMOsqHsY',
  'i': 'EDB8gKNwzurf33pV2hsyGR9XFOmitDhc0LUzDamcU2JR',
  'rd': 'EKuAljh08_m1CXluTeogXxHHlX93iqpmVfjMZ1cufhR-',
  's': 'EB8MUTKRg8284YPduyhDgUVvNM192-247KcRaXhIxoni',
  'a':
  {
    'd': 'ELD17t_hP-FJFTV5Iev7awrNVB8uABFAn0_xVEFrIm-t',
    'u': '0ABVbaoAh9HG7yVStjrYzNQa',
    'i': 'EKr8JLtfqWCmHrxO3yu8ocS2n9o0Tlspeaqm9ZOf3FM1',
    'rd': 'EAbxluMtNsPnHFu9texttH3G6B1s3ZqmJwpLhNbgc6ib',
    'issued': '2026-10-24T09:30:00.000000+00:00',
    'expires': '2026-10-31T22:00:00.000000+00:00',
    'rc': {'myEmptySpace': ['all']}
  },
  'e':
  {
    'd': 'EJf-YBNcc-aJg2fN89mhmWWh_ZQ6jA3IKPxAE6yd7_hd',
    'u': '0AAzOwU4s4tz6NfWZwjVvqRx',
    'guardian':
    {
      'd': 'EOiJkANQejaXxO9YqPhGZKDo6TAoA8kPPWh5QrgJFpvG',
      'u': '0ABO77Mb-lpbomUCGZgJ4Htz',
      'n': 'EONzgE0pcYF7hz1E_VVN4ea9iv75Q_C2ifhsZV1j6uQp',
      's': 'EDJzXg34opr6-odFix6lpiccfHF7ksITHuIES_MQRH4b',
      'o': 'I2I'
    }
  },
  'r':
  {
    'd': 'EFPxq4WPl29szUqbrQIviOh_Ls_RlrYbp4L-fdQH0XrX',
    'l': ''
  }
}
```

# ACDC Blinded Registry Example

## Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


## Registry Placeholder Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "1",
  "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "dt": "2025-08-01T18:06:10.988921+00:00",
  "b": "ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| `d` | `ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J` | BLID (Blinding SAID) |
| `u` | `aG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv8` | UUID salty nonce blinding factor, random or HD generated |
| `td` | `1AAP` | Transaction ACDC SAID field value, top-level `d` |
| `ts` | `1AAP` | Transaction state value string |

Unblinded attribute block as attachment to disclosure of TEL Event


```
-aAYECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6JaG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv81AAP1AAP
```


Placeholder  updates provide herd privacy wrt correlation to date time of actual state changes

# ACDC Blinded Registry Example ...

## Registry Issued Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "2",
  "p": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "dt": "2020-08-02T12:00:20.000000+00:00",
  "b": "EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| `d` | `EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t` | Dummied BLID (Blinding SAID) |
| `u` | `aLfCdNAnc-0P2SiruarZSajXiUWu5iU2VfQahvpNCyzB` | UUID salty nonce blinding factor, random or HD generated |
| `td` | `EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M` | Transaction ACDC SAID field value, top-level `d` |
| `ts` | `0Missued` | Transaction state value string |

# ACDC Blinded Registry Example ...

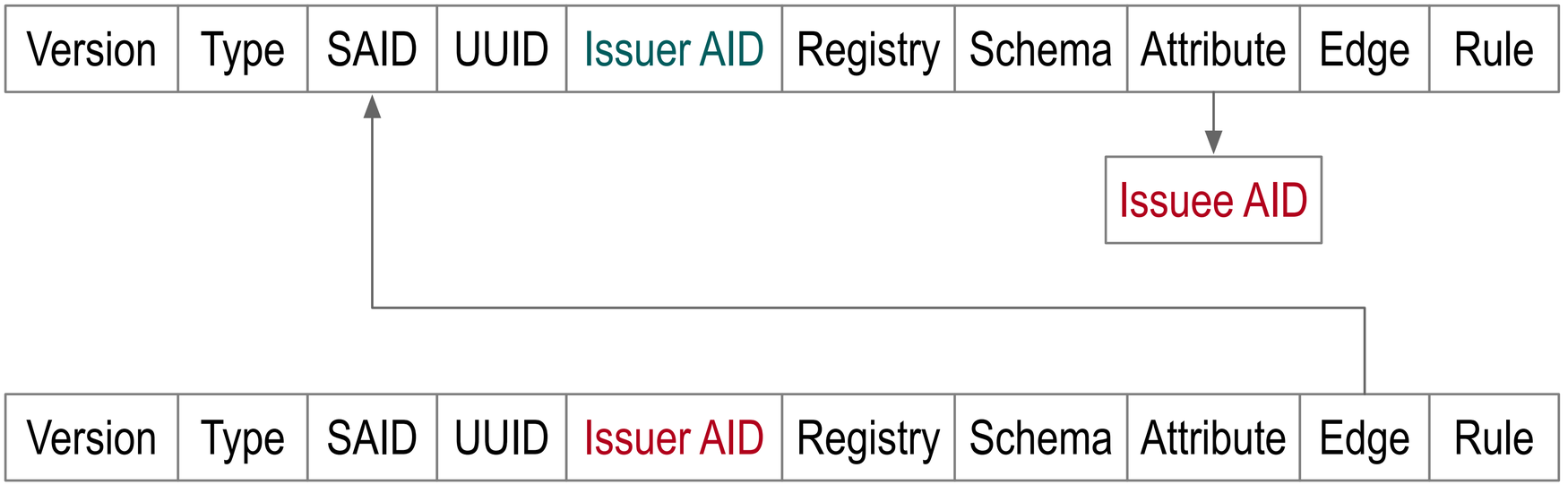
## Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EM8B1uDhWaJLfpIiEqgp-3EurGUcbfe7u2k5AarDl2XD",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| d | EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN | Dummied BLID (Blinding SAID) |
| u | aGx7b16vGHVPT56tX30kYOEzTwiVY4aabc4k9AawYyZG | UUID salty nonce blinding factor HD generated |
| td | EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M | Transaction ACDC SAID field value, top-level d |
| ts | Yrevoked | Transaction state value string |

# Reversion of Delegation

*Revocable delegated authority problem*:
When a delegatee's authority to issue authorizations is revoked, what is the status of any authorizations/entitlements issued before the authority was revoked?
How can a validator securely verify the status of prior valid issuances from a chain-of-authority when the chain is subsequently broken?



# Bound Blinded ACDC Registry Recovation

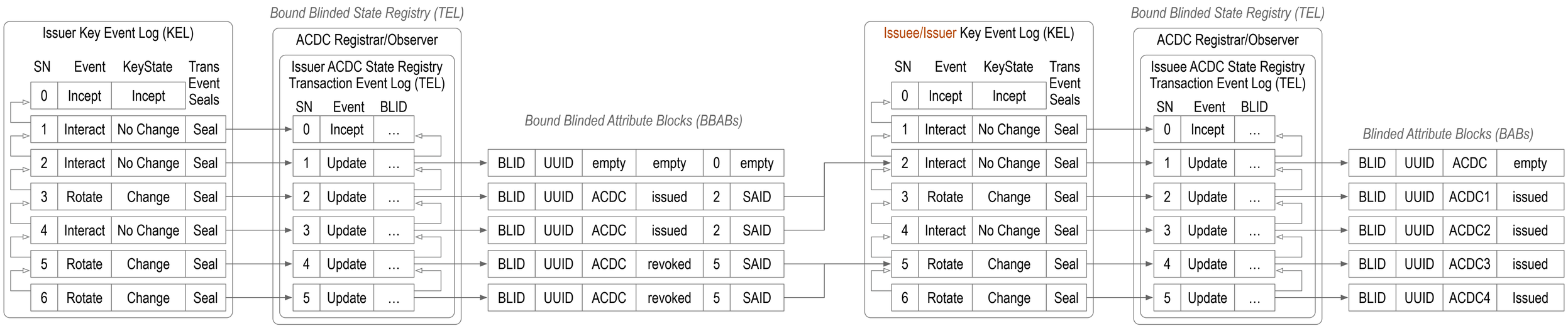
| Version | Type | SAID | UUID | Issuer AID | Registry | Schema | Attribute | Edge | Rule |
|---|---|---|---|---|---|---|---|---|---|

Issuee AID

| Version | Type | SAID | UUID | Issuer AID | Registry | Schema | Attribute | Edge | Rule |
|---|---|---|---|---|---|---|---|---|---|

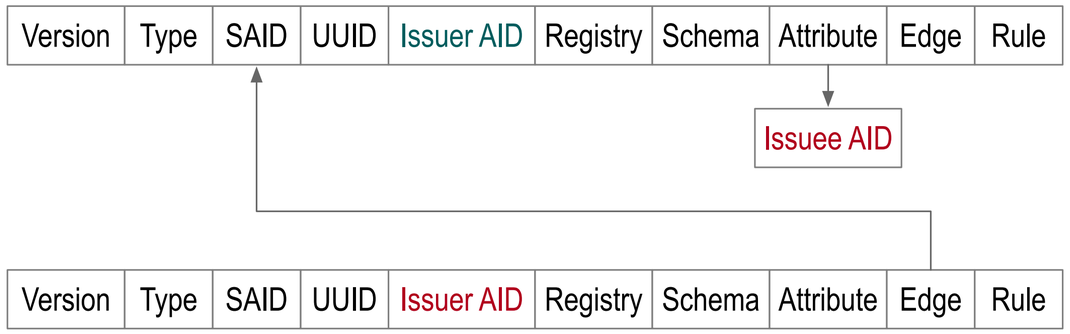
Bound Blinded Attribute Block (BBAB)

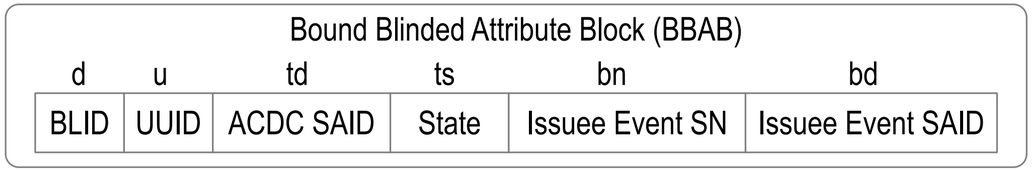
| d | u | td | ts | bn | bd |
|---|---|---|---|---|---|
| BLID | UUID | ACDC SAID | State | Issuee Event SN | Issuee Event SAID |

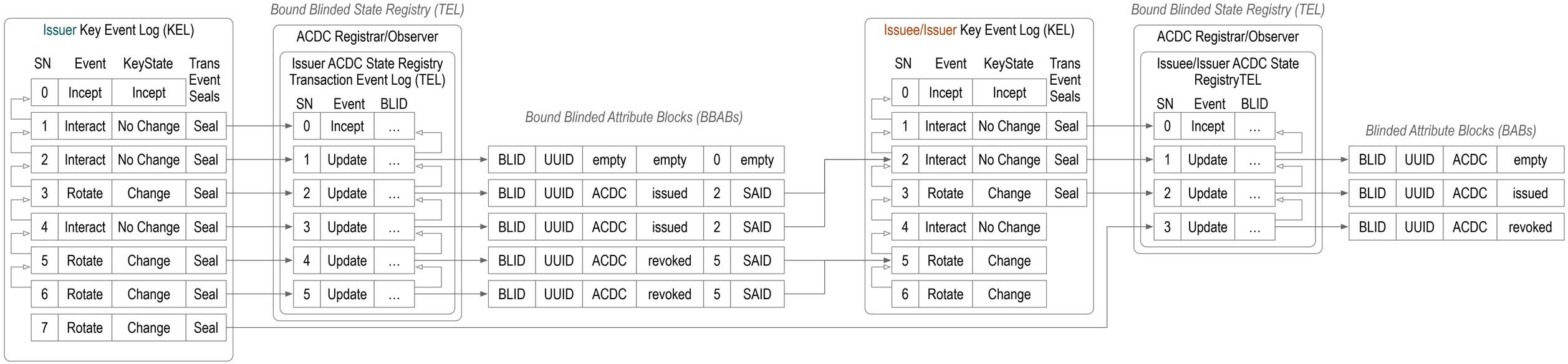



Breaking the chain-of-authority while preserving validity of prior valid issuances

# Bound Blinded ACDC Registry Authority Reversion



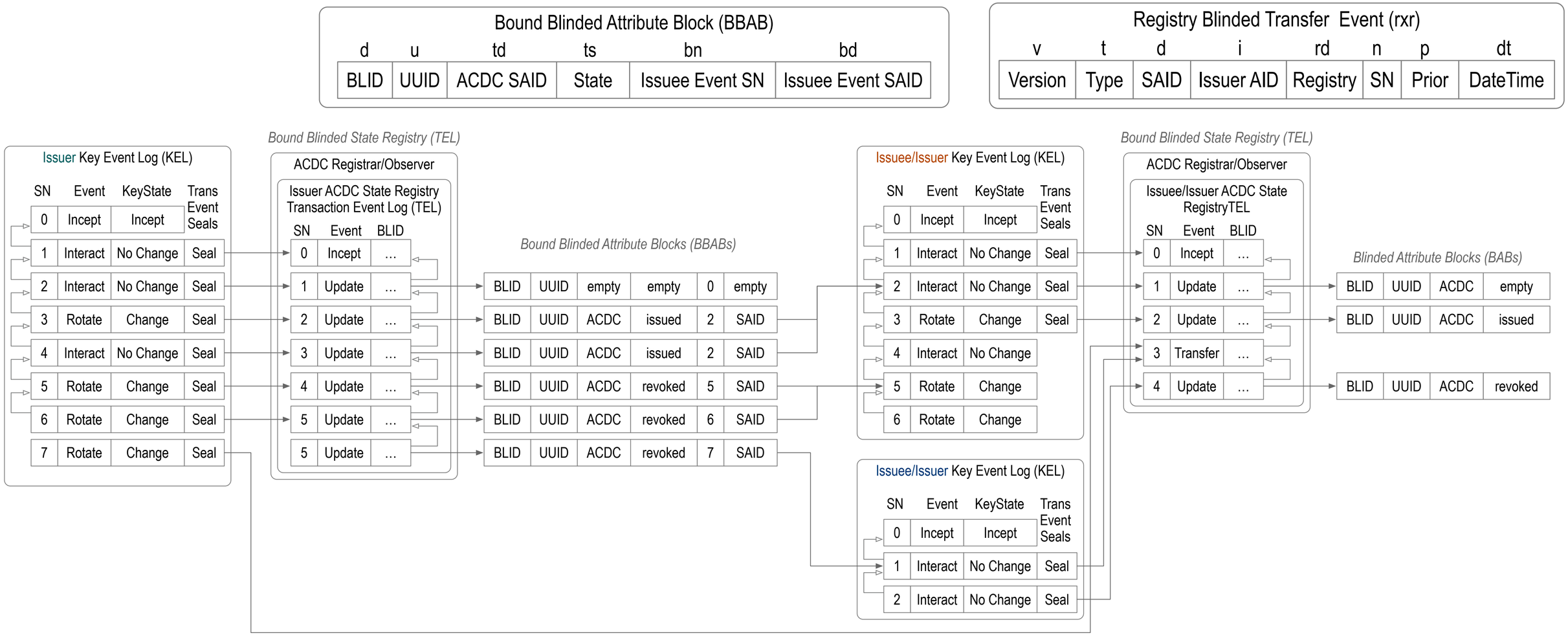

Reverting authority to delegator when delegatee's authority is revoked

# Transfer Registry

Replace the controller of registry without reverting the authority to delegator

Key rotation can already transfer control of AID and hence registry to new controllers

New registry transfer *xfr* event to transfer control of registry only.



# ACDC Bound Blinded Registry Example

## Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


## Registry Placeholder Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "1",
    "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "dt": "2025-08-01T18:06:10.988921+00:00",
    "b": "EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| d | EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c | BLID (Blinding SAID) |
| u | aJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa | UUID salty nonce blinding factor, HD generated |
| td | 1AAP | Transaction ACDC SAID field value, top-level d |
| ts | 1AAP | Transaction state value string |
| bn | MAAA | Bound Issuee key event sequence number field |
| bd | 1AAP | Bound Issuee key event SAID field |

## Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAaEFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0caJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa1AAP1AAPMAAA1AAP
```


Placeholder  updates provide herd privacy wrt correlation to date time of actual state changes

# ACDC Bound Blinded Registry Example …

## Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "2",
    "p": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "dt": "2020-08-02T12:00:20.000000+00:00",
    "b": "EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| d | EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4 | BLID (Blinding SAID) |
| u | aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTkO | UUID salty nonce blinding factor, HD generated |
| td | EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M | Transaction ACDC SAID field value, top-level d |
| ts | 0Missued | Transaction state value string |
| bn | MAAB | Bound Issuee key event sequence number field |
| bd | EJOnAKXGaSyJ_43kit0V806NNeGWS07lfjybB1UcfWsv | Bound Issuee key event SAID field |

## Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTkOEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M0MissuedMAABEJOnAKXGaSyJ_43kit0V806NNeGWS07lfjybB1UcfWsv
```

# ACDC Bound Blinded Registry Example …

## Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EDXGqM-V_sfj65Dk-lgMnpOZ9DYziqMaYkrju7NRKpFn",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv"
}
```


| Virtual Label | Value | Description |
|---|---|---|
| d | EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv | BLID (Blinding SAID) |
| u | aFudXE-d0b2owzZNBjd78sx4kCTJx-RTP_Zd19HRUcVD | UUID salty nonce blinding factor, HD generated |
| td | EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M | Transaction ACDC SAID field value, top-level d |
| ts | Yrevoked | Transaction state value string |
| bn | MAAI | Bound Issuee key event sequence number field |
| bd | EDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk | Bound Issuee key event SAID field |

## Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGvaFudXE-d0b2owzZNBjd78sx4kCTJx-
RTP_Zd19HRUcVDEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4MYrevokedMAAIEDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk
```